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A retrospective study on the efficacy of prenatal diagnosis for pregnancies at risk of mitochondrial DNA disorders

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PURPOSE: Prenatal diagnosis of mitochondrial DNA (mtDNA) disorders is challenging due to potential instability of fetal mutant loads and paucity of data connecting prenatal mutant loads to postnatal observations. Retrospective study of our prenatal cohort aims to examine the efficacy of prenatal diagnosis to improve counseling and reproductive options for those with pregnancies at risk of mtDNA disorders.

METHODS: We report on a retrospective review of 20 years of prenatal diagnosis of pathogenic mtDNA variants in 80 pregnant women and 120 fetuses.

RESULTS: Patients with undetectable pathogenic variants ($n = 29$) consistently had fetuses free of variants, while heteroplasmic women ($n = 51$) were very likely to transmit their variant (57/78 fetuses, 73%). In the latter case, 26 pregnancies were terminated because fetal mutant loads were >40%. Of the 84 children born, 27 were heteroplasmic (mutant load <65%). To date, no medical problems related to mitochondrial dysfunction have been reported.

CONCLUSION: Placental heterogeneity of mutant loads questioned the reliability of chorionic villous testing. Fetal mutant load stability, however, suggests the reliability of a single analysis of amniotic fluid at any stage of pregnancy for prenatal diagnosis of mtDNA disorders. Mutant loads under 40% reliably predict lack of symptoms in the progeny of heteroplasmic women.

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INTRODUCTION

Oxidative phosphorylation disorders are among the most common causes of inborn errors of metabolism. At least 20% of them are ascribed to pathogenic mitochondrial DNA (mtDNA) variants.[1] Unlike the nuclear genome, the mtDNA genome is strictly maternally inherited and located within the mitochondrial matrix. The large number of mtDNA molecules per cell allows for heteroplasmy—that is, coexistence of wild-type and mutant mtDNA species defining a mutant load (% of mutant mtDNA molecules versus total number of mtDNA copies). Mitotic segregation of mtDNA molecules is responsible for broad spatiotemporal variations in mutant loads across tissues. Patients exhibit disease symptoms when the mutant load reaches a certain threshold (usually >60%), and mtDNA mutations are associated with a broad clinical spectrum.[2–6] Clinical manifestations can indeed be restricted to a single affected tissue (Leber hereditary optic neuropathy, OMIM 535000) but multisystemic organ involvement including the central nervous system is common and frequently life threatening.[7] Among these severe diseases, three syndromes are particularly frequent: neurogenic muscle weakness, ataxia, retinitis pigmentosa (NARP, OMIM551500); mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke-like episodes (MELAS, OMIM54000); and Leigh syndromes (OMIM256000).

Two main classes of pathogenic mtDNA variants have been reported—namely, extensive sporadic rearrangements (deletions or duplications) and single-nucleotide pathogenic variants. The former are mostly sporadic and the latter are responsible for a wide range of diverse maternally inherited conditions.

Couples at risk of transmitting pathogenic mtDNA variants often seek reproductive advice but prenatal diagnosis (PND) of mtDNA mutations remains challenging. Based on the assessment of mutant loads in prenatal sample(s), PND aims to predict the risk of the fetus being severely affected after birth.[8–13] The choice of the embryonic or fetal material to analyze is questionable. While chorionic villus sampling (CVS) can be performed very early in pregnancy, the topographic distribution of pathogenic mtDNA variants in the placenta can be unpredictable, limiting its value for PND.[13] Amniotic fluid samples (AFS) may be more representative of mutant load within the fetus but are taken later during pregnancy, adding time constraints when a termination of pregnancy has to be considered. Whether mutant load fluctuates during pregnancy is also unclear, and therefore, the optimal stage of pregnancy to perform PND remains debated. Finally, published data connecting prenatal mutant loads to postnatal observations are limited, making the predictive power of prenatal testing uncertain. To add a level of complexity, these parameters might vary with mtDNA variants. More clinical experience is therefore critical to resolve these issues.

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Despite these uncertainties, we decided 20 years ago to offer prenatal diagnosis to at-risk couples to help reduce their risk of having a severely affected child. Here we report on observed outcomes of 120 at-risk pregnancies of women referred to our center over the last two decades (1999–2019), relating these to the findings of prenatal genetic testing for pathogenic mtDNA variants.

MATERIALS AND METHODS

A retrospective study was undertaken to evaluate the clinical usefulness of prenatal diagnosis for mtDNA disorders. A total of 80 pregnant women were included in the study. They were either heteroplasmic patients or at risk for mtDNA disorders and were referred to the Hôpital Necker-Enfants Malades medical genetics clinic for genetic counseling over the last 20 years (see Table 1). Their ages at time of pregnancy ranged from 17 to 41 years (mean: 30 years). Pregnant women each underwent 1–6 rounds of PND, and 120 fetuses at risk for inheriting a pathogenic mtDNA variant were tested. Of these 120 fetuses, 38 were previously reported in the literature.[8–10,13]

Couples requested prenatal diagnosis because a pathogenic mtDNA variant had previously been identified in the pregnant woman or close maternal relative(s). All patients were referred to a clinical geneticist to be informed about the limits and positive and negative predictive values of prenatal testing. They were advised that a first prenatal test could be performed on CVS but given mutant load fluctuations across fetal tissues, the results might be confirmed on a second sample from another tissue. Furthermore, due to possible variation of mutant loads during in utero development, a third evaluation of fetal mutant load should be recommended later during pregnancy (third trimester).

For all but six patients, pathogenic mtDNA variants were found within mitochondrial transfer RNA (tRNA) (*MT-TL1*, *MT-TN*, or *MT-TK*; 28 patients) or protein-coding (*MT-ATP6*, *MT-ND1*, *MT-ND3*, *MT-ND5*, *MT-ND6*, *MT-CO2*, *MT-CYB*; 46 patients) genes. Testing covered 17 pathogenic variants, all but 2 (m.15108T>G in the *MT-CYB* gene and m.7965T>C in the *MT-CO2* gene) having previously been reported as pathogenic. The most common of these 17 were the *MT-ATP6* (m.8993T>G/C and m.9185T>C; 24 patients) and *MT-TL1* (m.3243A>G; 26 patients) variants.

To evaluate the predictive value of maternal mutant loads, mtDNA variants were assessed in white blood cells and, when possible, urinary pellets or oral mucosa (Table 1). Paternal white blood cell DNA was systematically tested to rule out contamination by maternal DNA during the PND procedure. All patients gave written informed consent for prenatal sampling and genetic analyses. Parental informed consent was obtained for fetal analyses.

CVS was performed at 12 weeks of amenorrhea. AFS were collected at 16 weeks (AFS1) or 32 weeks (AFS2). For 92 of the 120 pregnancies, only one procedure was conducted (18 CVSs, 73 AFS1s, and 1 AFS2). In 21 cases, both CVS and AFS1 were performed. In two pregnancies, both AFS1 and AFS2 samples were collected and in five cases, all three procedures (CVS, AFS1, and AFS2) were carried out, to rule out putative variations in mutant loads during in utero development. In two pregnancies, a PND was performed to confirm a preimplantation genetic diagnosis (PGD, fetuses 54A and 31C; see Table 1). When possible, prenatal results were confirmed using cord blood sampled at birth (29/84 children)—or fetal tissues, in the case of termination of pregnancy (18/26 fetuses).

Mutant load quantification

DNA was extracted by using classical phenol extraction method,[14] or the Chemagic Prime DNA Blood kit, PerkinElmer, Villebon-sur-Yvette, France. Mutant loads were assessed via semiquantitative fluorescent polymerase chain reaction (PCR) tests with restriction enzyme digestion.[8,10,15] These methods are much more dated than others such as next-generation sequencing technologies but they are highly sensitive and accurate, very fast, and they require a very small amount of starting DNA.[10,15] Primers were chosen using oligo 6.0 software (Medprobe, Oslo, Norway). Cross-hybridization of oligonucleotide primers with genomic DNA was ruled out by PCR amplification using mtDNA-less ρ° cells.[16] A limited number of PCR cycles was performed to minimize the risk of underestimating mutant load as a result of heteroduplex formation during the reaction.

After digestion, PCR products (1 µL) were electrophoresed on an ABI 3130xl 24-capillary sequencer (Life Technologies, Carlsbad, CA, USA). Results were analyzed with GeneMapper software (Life Technologies). The mutant load was calculated by dividing the mutant peak area by the sum of the wild-type and mutant peak areas. Each assay was carried out in duplicate or triplicate. For all mtDNA variants, the estimated mutant load detection threshold was 2%.[8,10,15] The estimated interassay variation was also 2%.[10,15] For the m.3243A>G variant, the blood mutant load was corrected by maternal age, in accordance with Grady et al.;[17] age-adjusted blood level blood heteroplasmy/$0.977^{(age\ +\ 12)}$.

RESULTS

Couples ($n = 80$) were offered PND when a known or probable pathogenic mtDNA variant or a mtDNA deletion had been identified in the pregnant woman (51/80) or a close maternal relative (29/80; see Table 1). Women carrying no detectable mtDNA variant in white blood cells, urinary pellets, or oral mucosa DNA were still offered PND (for a total of 41 pregnancies) because they were closely related to affected patients (e.g., siblings, previous children, nieces or nephews). The remaining 51 women (79 pregnancies) were offered PND because they carried pathogenic mtDNA variants in their circulating leukocytes. For all but six patients, the pathogenic mtDNA variant was a single-nucleotide substitution. PND was performed on DNA extracted from CVS (44 pregnancies), AFS1 (101 pregnancies), or AFS2 (8 pregnancies) samples. For 28 pregnancies, tests were repeated at various gestational stages to exclude putative variations of fetal mutant loads during development.

Segregation of pathogenic mtDNA variants during pregnancy

Reliable quantification of pathogenic mtDNA variant loads was possible for 119 of the 120 fetuses; for the remaining fetus (64B), maternal DNA contamination hampered PND. A wide distribution of mutant loads was observed between the fetuses, ranging from 0% to 100%. Among the aforementioned 119 fetuses, 62 (52%)—including 41 borne by women with undetectable pathogenic mtDNA variants but closely related to heteroplasmic patients—exhibited no pathogenic variants, while the other 57 (48%) had either heteroplasmic or homoplasmic pathogenic variants.

The risk of heteroplasmic women to pass on the mtDNA variant was therefore 73%, (57/78 fetuses being heteroplasmic). These 57 fetuses were classified into three groups based on their mutant loads, placing them at low (<30%), intermediate (30–60%), or high (>60%) risk of developing a severe mtDNA disorder. Among these fetuses, 21 were considered at low, 15 at high, and 21 at intermediate risk (where no clear genotype–phenotype correlation is possible). The m.3243A>G variant was the most frequent genotype associated with this gray zone of intermediate risk. It is also worth noting that fetal mutant loads were consistently <80% for the m.3243A>G genotype, while pathogenic variants of respiratory chain subunit coding genes were mainly clustered in the low- or high-risk groups: only 5 of the 39 fetuses concerned belonged to the intermediate-risk group (Fig. 1a).

Value of AFS analysis in PND of pathogenic mtDNA variants

To detect potential variations of mutant loads across fetal tissues, both CVS and AFS procedures were conducted for 26 of the 120 cases, and fetal tissues were collected after autopsy in 18 cases. Mutant loads were identical for both procedures in all but two fetuses (43A and 68A), which carried the m.3243A>G and m.9185T>C variants respectively. Compared with AFS results, CVS for these two fetuses revealed either lower (fetus 43A, −16%) or higher (fetus 68A, +25%) mutant loads. Analysis of cord blood for fetus 68A turned up pathogenic mtDNA variant levels similar to those for AFS, but lower than with CVS. In fetus 40A, the mutant load was 48% for CVS and 32% for cord blood. In the last two cases, (fetuses 40A and 68A) significant differences in pathogenic mtDNA variant load were found in multiple placenta biopsies of

Table 1. Prenatal diagnosis of mtDNA mutations.

Mother	Fetus	Maternal age (years)	Reason for prenatal diagnosis	Gene	Maternal pathogenic variant[a] and heteroplasmy levels (%)					Fetal mutant loads (%)			Cord blood levels (%)	Mutant loads in fetal tissues (mean +/-SD%, *n* = number of tissues)	Outcome	Child age (years)	Publications in which patient was previously described
						Blood	Urine	Oral cells	Other	CVS	AFS1	AFS2					
1	1A	37	Previously affected child	*MT-TL1*	m.3243A>G	0	0		0 (hair)	0	0		0		Healthy at 3 years	13	Bouchet et al.,[8] family 3
2	2A	29	Affected brother	*MT-TL1*	m.3243A>G	0	0	0		0	0				Healthy at 6 years	15	Bouchet et al.,[8] family 2
	2B	35		*MT-TL1*	m.3243A>G						0				Pregnancy continued	9	N
3	3A	30	Affected nephew	*MT-TL1*	m.3243A>G	0	0			0	0				Male baby healthy at 8 years	17	Bouchet et al.,[8] family 1
	3B	33		*MT-TL1*	m.3243A>G						0				Female baby healthy at 5 years	14	Bouchet et al.,[8] family 1
	3C	38		*MT-TL1*	m.3243A>G						0				Female baby healthy at birth	9	N
4	4A	43	Affected sister	*MT-TL1*	m.3243A>G	0	0				0				Male baby healthy at birth	4	N
5	5A	29	Affected sister	*MT-TL1*	m.3271T>C	0					0		0		Pregnancy continued	2	N
6	6A	28	Previously affected child	*MT-ND1*	m.3697G>A				0 (m)		0				Healthy at 3 years	5	N
	6B	31		*MT-ND1*	m.3697G>A						0				Pregnancy continued	2	N
7	7A	37	Affected sister	*MT-TN*	m.5703G>A	0	0			0					Pregnancy continued	3	N
8	8A	40	Previously affected child	*MT-ATP6*	m.8993T>G	0				0	0				Pregnancy continued	17	Steffann et al.[9]
	8B	46		*MT-ATP6*	m.8993T>G						0				Trisomy 21 (TOP)		Steffann et al.[9]
9	9A	31	Affected nephew	*MT-ATP6*	m.8993T>G	0					0				Pregnancy continued	16	Steffann et al.[9]
	9B	33		*MT-ATP6*	m.8993T>G						0				Pregnancy continued	14	Steffann et al.[9]
	9C	36		*MT-ATP6*	m.8993T>G						0				Pregnancy continued	11	Steffann et al.[9]
10	10A	35	Previously affected child	*MT-ATP6*	m.8993T>G	0		0			0				Pregnancy continued	3	N
11	11A	35	Previously affected child	*MT-ATP6*	m.8993T>G	0					0		0		Male baby healthy at birth	<1	N
12	12A	39	Previously affected child	*MT-ATP6*	m.8993T>G	0	0				0				Ongoing pregnancy		N
13	13A	29	Affected niece	*MT-ATP6*	m.9185T>C	0	0				0				Pregnancy continued	2	N
14	14A	37	Previously affected child	*MT-ND3*	m.10158T>C	0	0	0			0				Male child healthy at 15 years	15	N
15	15A	34	Previously affected child	*MT-ND3*	m.10191T>C	0	0	0			0				Healthy at 2 years	10	N
	15B	37		*MT-ND3*	m.10191T>C					0					Pregnancy continued	7	N

Table 1 continued

Mother	Fetus	Maternal age (years)	Reason for prenatal diagnosis	Gene	Maternal pathogenic variant[a] and heteroplasmy levels (%)					Fetal mutant loads (%)			Cord blood levels (%)	Mutant loads in fetal tissues (mean +/-SD%, *n* = number of tissues)	Outcome	Child age (years)	Publications in which patient was previously described
						Blood	Urine	Oral cells	Other	CVS	AFS1	AFS2					
16	16A	34	Previously affected child	*MT-ND3*	m.10191T>C	0					0	0			Female baby healthy at birth	14	N
	16B	39		*MT-ND3*	m.10191T>C						0				Pregnancy continued	7	N
17	17A	28	Affected sister	*MT-ND3*	m.10191T>C	0	0				0		0		Female baby healthy at birth	5	N
	17B	31		*MT-ND3*	m.10191T>C						0		0		Female baby healthy at birth	2	N
18	18A	32	Affected nephew	*MT-ND5*	m.12706T>C	0	0				0				Female baby healthy at birth	5	N
19	19A	34	Previously affected child	*MT-ND5*	m.12706T>C	0	0				0		0		Female baby healthy at birth	1	N
20	20A	43	Previously affected child	*MT-ND5*	m.13513G>A	0	0	0			0	0	0		Female baby healthy at birth	6	N
21	21A	27	Affected mother	*MT-ND5*	m.13513G>A	0	0				0				Ongoing pregnancy		N
22	22A	42	Previously affected child	*MT-ND5*	m.13513G>A	0	0				0				Pregnancy continued	6	N
23	23A	29	Previously affected child	*MT-ND5*	m.13513G>A	0	0				0				Pregnancy continued	4	N
24	24A	26	Affected nephew	*MT-ND6*	m.14459G>A	0					0				Female baby healthy at birth	1	N
25	25A	36	Previously affected child	*MT-CYB*	m.15108T>G	0	0				0				Pregnancy continued	1	N
26	26A	39	Previously affected child		m.(599_?)_(?_16193)del	0					0				Pregnancy continued	11	N
27	27A	33	Previously affected child		m.(599_?)_(?_16193)del	0					0				Pregnancy continued	8	N
	27B	35			m.(599_?)_(?_16193)del	0					0				Pregnancy continued	10	N
28	28A	39	Previously affected child		m.(599_?)_(?_16193)del	0					0				Pregnancy continued	9	N
29	29A	30	Previously affected child		m.(599_?)_(?_16193)del	0					0				Pregnancy continued	8	N
	29B	34			m.(599_?)_(?_16193)del	0					0				Pregnancy continued	4	N
30	30A	29	Affected brother	*MT-TL1*	m.3243A>G	5	35	10	5 (hair)	20	15	15			Male child healthy at 5 years	10	Monnot et al.,[10] patient 5
	30B	30		*MT-TL1*	m.3243A>G						70			71+/-3 (*n* = 3)	TOP		Monnot et al.,[10] patient 5

Table 1 continued

Mother	Fetus	Maternal age (years)	Reason for prenatal diagnosis	Gene	Maternal pathogenic variant[a] and heteroplasmy levels (%)					Fetal mutant loads (%)			Cord blood levels (%)	Mutant loads in fetal tissues (mean +/-SD%, *n* = number of tissues)	Outcome	Child age (years)	Publications in which patient was previously described
						Blood	Urine	Oral cells	Other	CVS	AFS1	AFS2					
	30C	32		*MT-TL1*	m.3243A>G					4			2		Male child healthy at 3 years	8	Monnot et al.,[10] patient 5
31	31A	36	Previously affected child	*MT-TL1*	m.3243A>G	20	30	27		77				74+/-0 (*n* = 4)	TOP		Monnot et al.,[10] patient 2
	31B	36		*MT-TL1*	m.3243A>G					59				42+/-1 (*n* = 4)	TOP		Monnot et al.,[10] patient 2
	31C	38		*MT-TL1*	m.3243A>G						4		4		Male child healthy at 8 years	8	Monnot et al.,[10] patient 2
32	32A	23	Affected brother	*MT-TL1*	m.3243A>G	34	80	45	40 (hair)	35	33	29			Healthy at 2 years	15	Bouchet et al.,[8] family 4
	32B	25		*MT-TL1*	m.3243A>G					31	23	26			Pregnancy continued	13	Bouchet et al.,[8] family 4
33	33A	27	Affected brother	*MT-TL1*	m.3243A>G	21				60	63				TOP		Bouchet et al.,[8] family 5
	33B	27		*MT-TL1*	m.3243A>G					0	0				Healthy at 6 years	18	Bouchet et al.,[8] family 5
	33C	30		*MT-TL1*	m.3243A>G					0	0				Healthy at 1 year	15	Bouchet et al.,[8] family 5
34	34A	32	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	30	80	50		32	28				Pregnancy continued	10	Monnot et al.,[10] patient 6
35	35A	35	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	30	70			79				78+/-1 (*n* = 6)	TOP		Monnot et al.,[10] patient 7
36	36A	32	Affected sister	*MT-TL1*	m.3243A>G	7	15	8			0		0		Female baby healthy at 1 year	11	Monnot et al.,[10] patient 4
	36B	33		*MT-TL1*	m.3243A>G						60			63+/-1 (*n* = 5)	TOP		Vachin et al.,[13] fetus 2
37	37A	28	Affected nephew	*MT-TL1*	m.3243A>G	20	50	40		17	16		22		Male baby healthy at 9 years	9	Monnot et al.,[10] patient 3
	37B	32		*MT-TL1*	m.3243A>G						22		24		Male baby healthy at 4 years	4	N
38	38A	39	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	15	74	30		5	8		8		Male baby healthy at birth	7	N
39	39A	28	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	10		10		65					TOP		N
40	40A	26	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	40	55	62		48			32		Male baby healthy at birth	8	Vachin et al.,[13] fetus 5
41	41A	27	Affected brother	*MT-TL1*	m.3243A>G	30	60				0		0		Female baby healthy at 7 years	7	N
	41B	31		*MT-TL1*	m.3243A>G					0			0		Female baby healthy at 4 years	4	N
	41C	35		*MT-TL1*	m.3243A>G					42	40				TOP		N
42	42A	23	Affected sister	*MT-TL1*	m.3243A>G	23	35				41		44		Female baby	4	N

Table 1 continued

Mother	Fetus	Maternal age (years)	Reason for prenatal diagnosis	Gene	Maternal pathogenic variant[a] and heteroplasmy levels (%)					Fetal mutant loads (%)			Cord blood levels (%)	Mutant loads in fetal tissues (mean +/-SD%, *n* = number of tissues)	Outcome	Child age (years)	Publications in which patient was previously described
						Blood	Urine	Oral cells	Other	CVS	AFS1	AFS2					
															healthy at 1 year		
	42B	25		*MT-TL1*	m.3243A>G						30		35		Male baby healthy at birth	2	N
43	43A	26	Affected brother	*MT-TL1*	m.3243A>G	43	55	51		37	53				TOP		N
44	44A	22	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	4	5				0				Pregnancy continued	2	N
	44B	23		*MT-TL1*	m.3243A>G						0				Pregnancy continued	1	N
45	45A	25	Affected sister	*MT-TL1*	m.3243A>G	28					53			50+/-7 (*n* = 7)	TOP		N
	45B	27		*MT-TL1*	m.3243A>G						12		12		Pregnancy continued	4	N
46	46A	21	Affected nieces	*MT-TL1*	m.3243A>G	25	55				50				Pregnancy continued	5	N
	46B	26		*MT-TL1*	m.3243A>G						45		37		Female baby healthy at birth	<1	N
47	47A	33	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	33	62				36			38+/-4 (*n* = 5)	TOP		N
48	48A	29	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	42	64				62		64		Female baby healthy at 1 year	4	N
49	49A	33	Previously affected child	*MT-TL1*	m.3243A>G	12	52	22			0				Pregnancy continued	3	N
50	50A	31	Mother known to be heteroplasmic	*MT-TL1*	m.3243A>G	10					0		0		Pregnancy continued	2	N
51	51A	38	Previously affected children	*MT-CO2 (VUS)*	m.7965T>C	65				0	0				Female baby healthy at birth	7	N
52	52A	32	Affected brother	*MT-TK*	m.8344A>G	50	56				7				Ongoing pregnancy		
53	53A	37	Previously affected child	*MT-ATP6*	m.8993T>G	55				64	65			62+/-3 (*n* = 4)	TOP		Steffann et al.[9]
	53B	41		*MT-ATP6*	m.8993T>G						95				TOP		Vachin et al.,[13] fetus 8
54	54A	37	Previously affected child	*MT-ATP6*	m.8993T>G	17						0	0		Male child healthy at 10 years	13	Steffann et al.[9]
55	55A	30	Previously affected child	*MT-ATP6*	m.8993T>G	65				13	10	10			Pregnancy continued	18	Steffann et al.[9]
56	56A	28	Previously affected child	*MT-ATP6*	m.8993T>G	30				100					TOP		Steffann et al.[9]
57	57A	28	Affected brother	*MT-ATP6*	m.8993T>G	30					87			91+/-0 (*n* = 14)	TOP		Steffann et al.[9]
	57B	29		*MT-ATP6*	m.8993T>G						70			74+/-1 (*n* = 11)	TOP		Steffann et al.[9]
58	58A	25	Affected brother	*MT-ATP6*	m.8993T>G	75		80		30					Male child healthy at 14 years	14	Steffann et al.[9]
	58B	28		*MT-ATP6*	m.8993T>G					70	70			79+/-1 (*n* = 5)	TOP		Steffann et al.[9]

Table 1 continued

Mother	Fetus	Maternal age (years)	Reason for prenatal diagnosis	Gene	Maternal pathogenic variant[a] and heteroplasmy levels (%)					Fetal mutant loads (%)			Cord blood levels (%)	Mutant loads in fetal tissues (mean +/-SD%, *n* = number of tissues)	Outcome	Child age (years)	Publications in which patient was previously described
						Blood	Urine	Oral cells	Other	CVS	AFS1	AFS2					
	58C	30		*MT-ATP6*	m.8993T>G						94			92+/-6 (*n* = 5)	TOP		Vachin et al.,[13] fetus 7
	58D	30		*MT-ATP6*	m.8993T>G					25					Female baby healthy at 9 years	9	N
	58E	32		*MT-ATP6*	m.8993T>G					60	64				Female baby healthy at 8 years	8	N
	58F	36		*MT-ATP6*	m.8993T>G					14	12		17		Female baby healthy at 3 years	3	N
59	59A	26	Previously affected child	*MT-ATP6*	m.8993T>G	70	70	70		95				97+/-0 (*n* = 6)	TOP		N
60	60A	40	Previously affected child	*MT-ATP6*	m.8993T>G	7				0	0				Pregnancy continued	8	N
61	61A	27	Previously affected child	*MT-ATP6*	m.8993T>C	90				25					Pregnancy continued	8	N
62	62A	33	Previously affected child	*MT-ATP6*	m.8993T>G	18					0				Pregnancy continued	4	N
	62B	35		*MT-ATP6*	m.8993T>G						0				Pregnancy continued	2	N
63	63A	23	Previously affected child	*MT-ATP6*	m.8993T>G	4					0				Pregnancy continued	4	N
64	64A	32	Previously affected child	*MT-ATP6*	m.8993T>G	40	44				0				TOP (beta thalassemia)		N
	64B	33		*MT-ATP6*	m.8993T>G						und				TOP (beta thalassemia)		N
65	65A	30	Previously affected child	*MT-ATP6*	m.8993T>G	40				10	12	11			Male baby healthy at 20 years	20	N
66	66A	23	Affected brother	*MT-ATP6*	m.8993T>G	8					67				TOP		N
67	67A	36	Previously affected child	*MT-ATP6*	m.8993T>G	25					100			100 (*n* = 1)	TOP		N
68	68A	33	Previously affected child	*MT-ATP6*	m.9185T>C	21	14	17		55	25		25		Male baby healthy at birth	8	Vachin et al.,[13] fetus 9
69	69A	28	Affected sister	*MT-ATP6*	m.9185T>C	93	92				35		35		Female baby healthy at birth	3	N
70	70A	28	Affected brother	*MT-ATP6*	m.9185T>C	20					5		5		Female baby healthy at birth	4	N
71	71A	31	Previously affected children	*MT-ND3*	m.10191T>C	20	69				25				Ongoing pregnancy		N
72	72A	39	Previously affected children	*MT-ND3*	m.10197G>A	38				100				100 (*n* = 1)	TOP		Vachin et al.,[13] fetus 10
	72B	39		*MT-ND3*	m.10197G>A					100				100 (*n* = 1)	TOP		N
73	73A	17	Mother known to be heteroplasmic	*MT-ND3*	m.10197G>A	12					0		0		Pregnancy continued	7	N

Table 1 continued

Mother	Fetus	Maternal age (years)	Reason for prenatal diagnosis	Gene	Maternal pathogenic variant[a] and heteroplasmy levels (%)					Fetal mutant loads (%)			Cord blood levels (%)	Mutant loads in fetal tissues (mean +/-SD%, *n* = number of tissues)	Outcome	Child age (years)	Publications in which patient was previously described
						Blood	Urine	Oral cells	Other	CVS	AFS1	AFS2					
74	74A	28	Previously affected children	*MT-ND5*	m.13513G>A	8	29	10		0	0				Male baby healthy at birth	3	N
	74B			*MT-ND5*	m.13513G>A					50					Ongoing pregnancy		N
75	75A	39	Previously affected child	*MT-ND6*	m.14459G>A	21	67			50	50			51+/-2 (*n* = 7)	TOP		N
	75B	40		*MT-ND6*	m.14459G>A					98					TOP		N
76	76A	24	Previously affected child	*MT-ND6*	m.14459G>A	5					15		13		Female baby healthy at 1 year	1	N
	76B	26		*MT-ND6*	m.14459G>A						10				Ongoing pregnancy		N
77	77A	37	Previously affected child	*MT-ND6*	m.14487T>C	40					98			94+/-1 (*n* = 5)	TOP		N
78	78A	22	Affected brother	*MT-ND6*	m.14487T>C	7	21				0				Ongoing pregnancy		N
79	79A	32	Mother known to be heteroplasmic		m.(599_?)_(?_16193)del	HTP					0				Female baby healthy at birth	1	N
80	80A	28	Mother known to be heteroplasmic		m.(599_?)_(?_16193)del	HTP			HTP (m)		0				Female baby healthy at birth	5	N

AFS1 amniotic fluid sampled at 16 weeks of amenorrhea, *AFS2* amniotic fluid sampled at 32 weeks of amenorrhea, *CVS* chorionic villus sampling, *m* muscle, *mtDNA* mitochondrial DNA, *HTP* heteroplasmy, *N* no, *TOP* termination of pregnancy, *und* undetermined.

[a]NC_012920.1.

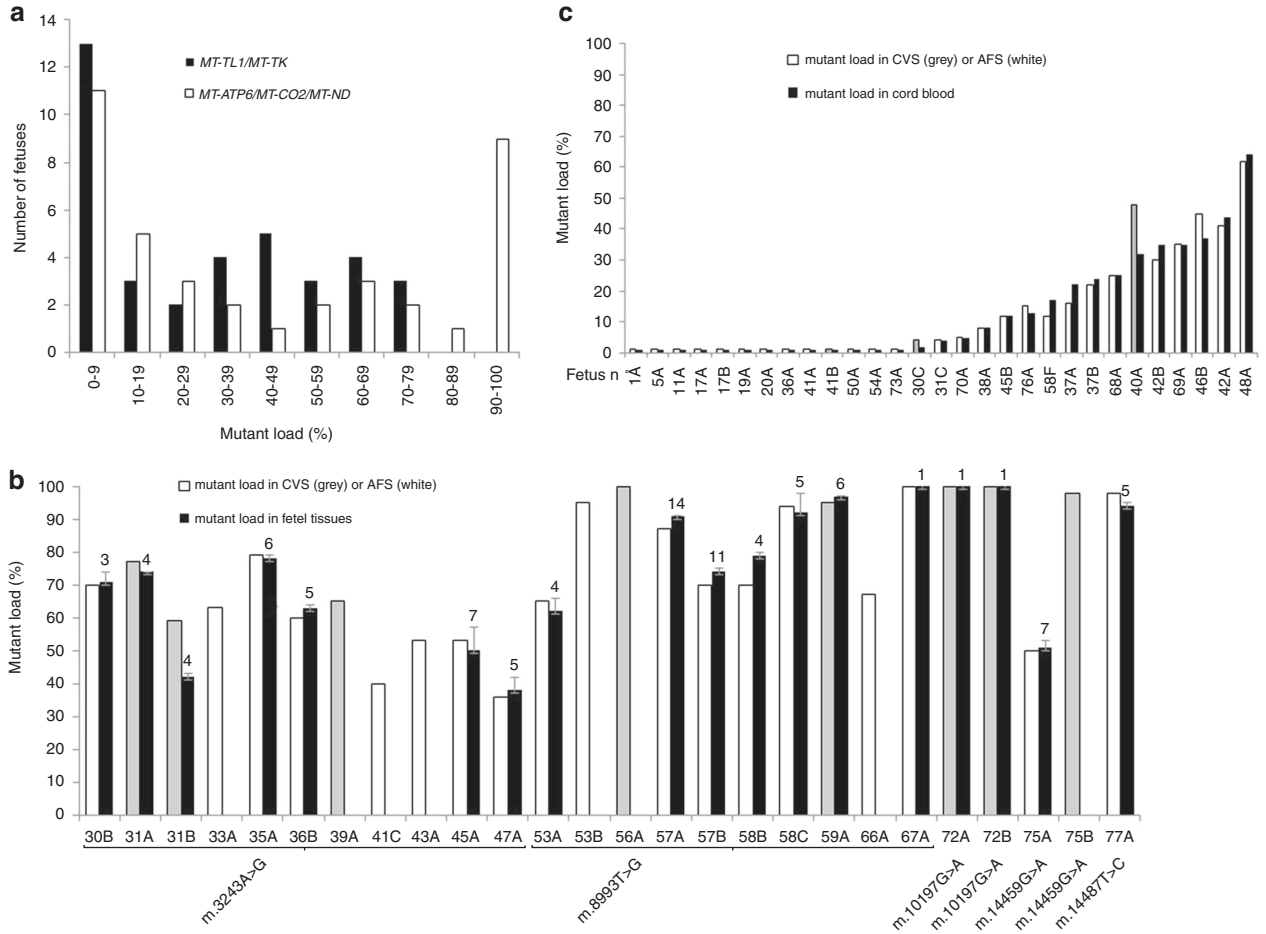


Fig. 1 Distribution of fetal mutant loads. (**a**) Distribution of fetal mutant loads according to type of mitochondrial DNA (mtDNA) mutation. Mutant loads were quantified for 76 fetuses of women carrying pathogenic mtDNA variants in mitochondrial transfer RNA (tRNA) (*MT-TL1*, and *MT-TK*; $n = 37$) or protein-coding (*MT-ATP6*, *MT-CO2*, and *MT-ND*; $n = 39$) genes. When more than one prenatal sample was available, mean fetal mutant load is shown. (**b**) Stability of pathogenic mtDNA variant loads across tissues in terminated pregnancies. In all, 26 pregnancies were terminated due to high fetal mutant loads, evaluated by amniotic fluid (AFS; white boxes) or chorionic villus (CVS; gray boxes) sampling. In 18 cases, analyses were carried out on fetal tissues (mean ± 1 SD, black boxes). The number of fetal tissues available for testing is indicated above each box. (**c**) Correlation between pre- and postnatal fetal mutant loads. Postnatal mutant loads were determined using cord blood sampled at birth for 29 children. Corresponding AFS (white boxes) or CVS mutant loads (gray boxes) are compared with cord blood levels (black boxes).

the same pregnant woman.[12] Clearly, these observations cast doubt on the value of CVS for PND of pathogenic mtDNA variants.

To further investigate possible variations of mutant loads across fetal tissues, autopsy specimens of 18 fetuses were analyzed (Fig. 1b). Fetal mutant loads were consistently similar (mean variation: 3%) to those found via AFS or CVS analysis, regardless of the pathogenic mtDNA variant. Taken together, our findings argue in favor of AFS for PND of pathogenic mtDNA variants and suggest mtDNA pathogenic variants are homogeneously distributed across fetal tissues.

Changes in mtDNA mutant load during fetal development

To detect potential fluctuations of mutant mtDNA load during antenatal development, both AFS1 and AFS2 samples were collected for 7 of the 120 pregnancies, and cord blood was sampled at birth for 29 of the 120 fetuses (Fig. 1c). No variation in mutant load was observed during fetal development, regardless of the pathogenic mtDNA variant (m.3243A>G, m.8993T>G, m.10191T>C, or m.13513G>A; see Table 1). Similarly, no significant difference between prenatal and cord blood mutant loads were noted (mean difference: 2%; Fig. 1c), except in fetus 40A, whose mutant load was markedly higher in the CVS sample (48%) than for cord blood (32%).[7] Furthermore, in two in vitro fertilization (IVF) pregnancies for which PGD was performed (fetuses 54A and 31C), cord blood mutant loads were similar to those found in the corresponding blastomeres. Altogether, these data point to the stability of mtDNA mutant loads throughout embryonic and fetal development, irrespective of the pathogenic mtDNA variant. Hence a single AFS mutant load evaluation at any stage of pregnancy is considered valid for PND of mtDNA disorders.

Predictive value of maternal mtDNA mutant load and age on pregnancy outcome

We drew on the data presented above in an attempt to estimate the predictive value of blood maternal mutant level for fetal mutant loads. A relatively close correlation between maternal and fetal mutant loads was observed for *MT-ND* variants ($R^2 = 0.7$, $p = 0.001$), despite clustering of fetal loads in either high or low levels of heteroplasmy (Fig. 2a). In contrast, poor correlations were noted for *MT-ATP6* ($R^2 = 0.08$; Fig. 2b) and m.3243A>G (*MT-TL1*, $R^2 = 0{\cdot}09$; Fig. 2c) variants. Correction of maternal mutant load according to age at sampling for the m.3243A>G variant increased the correlation (from $R^2 = 0.09$ to $R^2 = 0.11$). Finally, we questioned whether maternal age could affect pathogenic variant load,

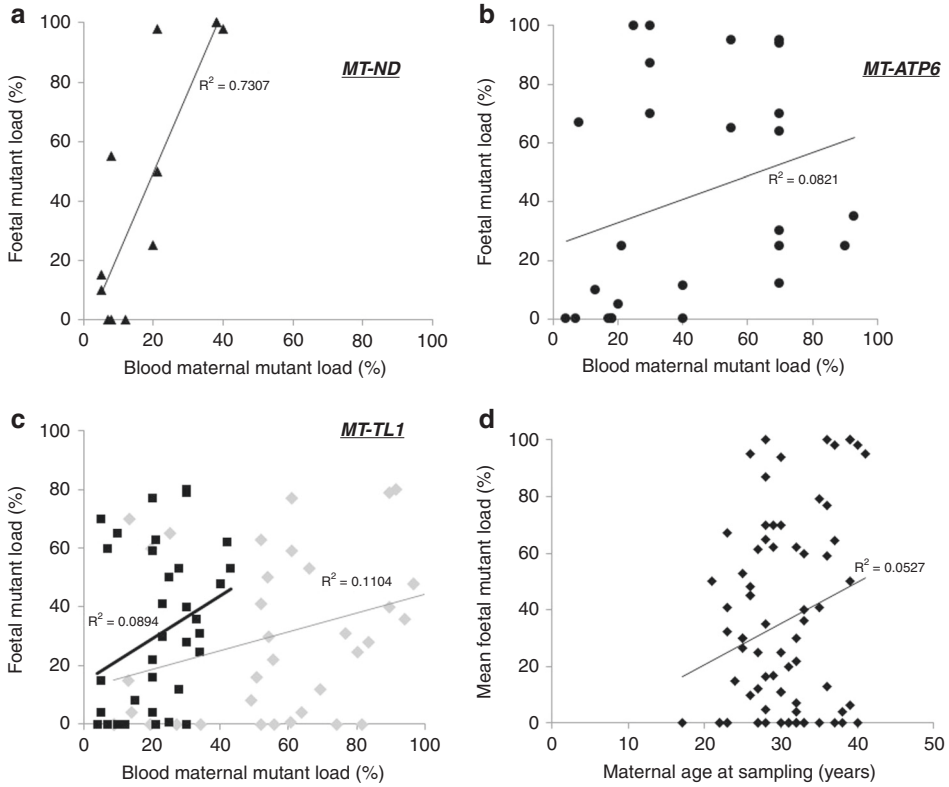


Fig. 2 Maternal factors and fetal mutant loads. (**a**–**c**) Correlation between maternal and fetal mutant loads. Maternal mutant loads were estimated using blood samples. Fetal amniotic fluid sample (AFS) mutant loads are given when available; in their absence, chorionic villus sample (CVS) mutant loads are used. For the m.3243A>G mutation, raw maternal mutant loads (black diamond) and age-corrected maternal mutant loads (gray diamonds), calculated in accordance with Grady et al.[17] are used. (**d**) Effect of maternal age on fetal mutant loads. Fetal mutant loads are plotted against maternal age at sampling. When several fetal mtDNA mutant loads were available, mean mutant load is used.

especially as clearance of mtDNA deletion and pathogenic variants has been reported in mouse oocytes.[18,19] Interestingly, a positive correlation between maternal age and fetal mutant load was observed in our cases ($R^2 = 0.05$, $p < 0.05$; Fig. 2d).

Impact of PND procedure on decision-making process

AFS fetal mutant loads exceeding 40% prompted the termination of 26 of the 120 pregnancies. Three other pregnancies were terminated for an unrelated genetic condition, and seven are ongoing. Of the 84 children born, 57 had no pathogenic mtDNA variants, and 27 had prenatal mutant loads under 65% (<30% in 18 cases and >30% in 9 remaining cases; see Table 1). For legal reasons, no further testing was performed after birth. French law (decree number 2000-570 of 23 June 2000) indeed states that genetic testing in asymptomatic minors can only be performed if the minor or his or her family can personally benefit from immediate preventive or curative measures. To date, no medical problems associated with a putative mitochondrial disorder have been reported (mean age: 7 years; range: 5 months to 20 years).

DISCUSSION

Here we review findings for 120 pregnancies in 80 women carrying, or at risk for, pathogenic mtDNA variants and referred to our center for PND over the last 20 years (1999–2019). For 29 of these women, pathogenic mtDNA variants were undetectable. They were still offered PND because they were closely related to an affected patient : low maternal mutant levels may escape detection, or mtDNA variants may be restricted to germline cells. None of these 29 women transmitted pathogenic mtDNA variants to their fetuses. Conversely to mutation-free patients, heteroplasmic patients have a high risk (73%) of transmitting the pathogenic variant to their progeny. In an attempt to predict likely pregnancy outcomes, we tried to correlate maternal and fetal mutant loads. Relatively close correlations between maternal and fetal mutant loads were observed, but a degree of complexity results from the variability of the mutant load across maternal postnatal tissues, especially for the m.3243A>G. Correcting the maternal blood mutant load by age, as suggested by Grady et al.,[17] increases the fetal/maternal correlation. The substantive variability in biologic transmission weakens the correlation, but our data corroborate previous ones showing that the risk of having a clinically affected child can be predicted on the basis of maternal m.3243A>G mutant load.[20]

Our study shows that about half of the fetuses (48%) carried homo- or heteroplasmic pathogenic variants of mitochondrial tRNA or protein-coding genes, the most common mtDNA variants concerning *MT-ATP6* and *MT-TL1*. Among them, 37% were considered at low risk, 26% at high risk, and 37% at intermediate risk, corresponding to a gray zone within which no clear correlation between genotype and phenotype is presently possible. On the basis of these results, nearly 22% of the pregnancies (26/120) were terminated due to CVS or AFS mutant loads exceeding 40%. High prenatal mutant loads (>60% to >80%, depending on variant) are considered strong predictors of serious disorders.[5,6,21] On the other hand, as postnatal mutant loads below 30% are usually associated with no or only mild disease, parents often opt to let pregnancy proceed when risk is low. A decision is harder to make when there is an intermediate risk (30% to 60%), especially since the m.3243A>G variant was the most frequent genotype in this category. Indeed, no data correlating intermediate prenatal mutant loads with postnatal outcomes are

presently available. More data are needed to establish correlations between mutant load and severity of disease in such situations, which is important to aid genetic counseling. Ultimately, 70% of the pregnancies were allowed to proceed, and 84 children were born. Of these 84 children, 57 were free of pathogenic mtDNA variants and 27 had mutant loads below 65%. Though there was no postnatal testing, children with heteroplasmic variant loads under 65% were followed up—over periods ranging from a few months to 20 years—and no further medical problems concerning potential mitochondrial disorders have been reported to date.

It is worth noting that highly variable mutant loads were observed for multiple pregnancies of women carrying pathogenic mtDNA protein-coding gene variants. Their fetuses had either very high (>90%) or very low (<10%) levels of pathogenic variants. A third of the heteroplasmic women failed to give birth to at least one healthy child, and most of these childless patients carried pathogenic variants in mitochondrial protein-coding genes. This might reflect extreme skewing for such variants. Indeed, several mtDNA variants, including m.8993T>G, are known to be more prone to extreme skewing.[22–25] Conversely, other pathogenic variants affecting mitochondrial tRNA genes were not detected at high levels by AFS analysis, suggesting counterselection of heavily mutated germline cells during oogenesis.

A practical aim of our study was to document how reliable CVS and AFS are, respectively, for PND. After performing both procedures for a subset of fetuses, we found they indicated identical mutant loads for all but two cases, in which inconsistencies were observed. In addition, significant variant load differences were occasionally observed in multiple placental biopsies.[13] Even if no consensus exists between different centers offering prenatal testing for pathogenic mtDNA variants, and most prenatal reports are based on CVS analysis, such inconsistencies question the reliability of CVS for PND of pathogenic mtDNA variants. In contrast, fetal mutant loads were the same in AFS samples and fetal tissue specimens obtained after termination of pregnancy, or in cord bloods sampled at birth, for all pathogenic variants. Thus, our findings support AFS for PND of pathogenic mtDNA variants and suggest mtDNA variants are homogeneously distributed across fetal tissues, as previously observed.[9,10,26–28] Our data also indicate the stability of mtDNA mutant loads throughout fetal development, irrespective of the pathogenic mtDNA variant. For this reason, we now consider that a single AFS analysis performed at any stage of pregnancy is acceptable for PND of mtDNA disorders. Our study gives some important data that should help guide reproductive choices for patients who seek advice to reduce their risk of having a severely affected child. As there are now more reproductive options for women who carry pathogenic mtDNA variants, a priority, and sometimes a challenge, is determining which option is most appropriate. Specialist reproductive advice is needed so that women/couples can make informed decisions about the reproductive option that is right for them. For women who are heteroplasmic and at high risk of transmitting the mtDNA variant, PGD may be considered an attractive option. PGD requires IVF-based procedures which are burdensome. Alternatives such as oocyte donation, or mitochondrial replacement therapy (MRT) might also be proposed. MRT is not permitted in France but is in the United Kingdom; furthermore, one birth has been reported following spindle transfer in Mexico.[29] Yet, MRT does not eliminate the residual risk of heteroplasmy as an accumulation of mutant mtDNA molecules has been reported in a limited number of cultured embryonic stem cells derived from the resulting oocytes and embryos, and in the United Kingdom, mitochondrial donation is offered in conjunction with prenatal diagnosis until more is known about the long-term outcomes of the technique.[30–32]

In this study, we show that prenatal diagnosis is an effective reproductive option for patients at risk of mtDNA disorder. Our results suggest there is a low level of risk for the progeny of pregnant women maternally related to heteroplasmic patients when no pathogenic mtDNA variants are detected in their white blood cells. More data are needed to determine the medical relevance of PND in such situations, and PGD may not be appropriate in such cases.[12] Conversely, fetuses of heteroplasmic women are likely to have mutant loads putting them at intermediate to high risk of mitochondrial disorders. For these women, our findings justify AFS analysis at any point in their pregnancies, since most (36/49) gave birth to healthy children. While all children born to date are healthy, heteroplasmic children should be subject to long-term follow up.

DATA AVAILABILITY

Protocols for quantification of each mtDNA variant and reagents are available on request.

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AUTHOR CONTRIBUTIONS

Conceptualization: J.-P.B., A.M., A.R. Data curation: J.B., R.B., S.M., J.S. J.-P.B., A.M. Formal analysis: M.M., Z.A., N.G., G.B. Investigation: Y.V., L.S., B.B., J.M. Methodology: J.S., S.M., A. M., J.-P.B. Writing—original draft: J.S, JP.B, A.M. Writing—review and editing: J.S., A.M.

COMPETING INTERESTS

The authors declare no competing interests.

ETHICS DECLARATION

This retrospective review has been approved by the Research Ethics Committee of the Necker-Enfants Malades University Hospital (IRB registration 00011928). All experiments were performed in accordance with relevant guidelines and French regulations. The Necker-Enfants Malades University Hospital has a general privacy statement informing patients that their data can be used for scientific research (https://www.aphp.fr/protection-des-donnees-personnelles).

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